

Numerical Solution of the Time-Fractional Burgers' Equation

December 2, 2025

1 Introduction

The Time-Fractional Burgers' Equation (FBE) is a nonlinear partial differential equation that models various physical phenomena, such as shock waves and fluid dynamics, but incorporates a memory effect through the Caputo fractional derivative. The governing equation is defined as:

$$\frac{\partial^\alpha u}{\partial t^\alpha} + au \frac{\partial u}{\partial x} - c \frac{\partial^2 u}{\partial x^2} = 0, \quad (0 < \alpha \leq 1) \quad (1)$$

where α is the fractional order of time differentiation, a is the nonlinear coefficient, and c is the viscous coefficient. The challenge in solving this equation numerically lies in addressing three main components:

1. The **Non-local (Memory) Fractional Derivative** ($\frac{\partial^\alpha u}{\partial t^\alpha}$).
2. The **Nonlinear Advection Term** ($au \frac{\partial u}{\partial x}$).
3. The **Viscous Term** ($c \frac{\partial^2 u}{\partial x^2}$).

1.1 Numerical Strategy

Our approach integrates three well-established numerical techniques to form a robust and efficient scheme:

1. **L1 Formula:** Approximates the Caputo fractional derivative, capturing the historical dependency (memory).
2. **Crank-Nicolson Scheme:** Provides an implicit, second-order accurate discretization for the spatial derivatives, ensuring excellent stability.
3. **Predictor-Corrector Method:** Linearizes the nonlinear advection term to enable direct solution via a linear system.

2 Discretization and Formulation

We discretize the domain into a grid: $x_j = j\Delta x$ and $t_n = n\Delta t$. We seek the numerical solution $u_j^n \approx u(x_j, t_n)$.

2.1 Approximation of the Fractional Derivative (L1 Formula)

The L1 formula provides a first-order approximation for the Caputo fractional derivative at time t_{n+1} :

$$D_t^\alpha u_j^{n+1} \approx K \left[(u_j^{n+1} - u_j^n) + \sum_{k=1}^n b_k (u_j^{n+1-k} - u_j^{n-k}) \right] \quad (2)$$

where $K = \frac{(\Delta t)^{-\alpha}}{\Gamma(2-\alpha)}$ and the memory weights are $b_k = (k+1)^{1-\alpha} - k^{1-\alpha}$.

2.2 Crank-Nicolson Discretization of Spatial Terms

The Crank-Nicolson method averages the spatial terms between the current time step (t_n) and the unknown future time step (t_{n+1}).

2.2.1 Viscous Term (Diffusion)

The linear diffusion term is discretized implicitly for stability:

$$cu_{xx} \approx \frac{c}{2} \left[\delta_x^2 u_j^{n+1} + \delta_x^2 u_j^n \right] \quad (3)$$

where $\delta_x^2 u_j = \frac{u_{j+1} - 2u_j + u_{j-1}}{(\Delta x)^2}$ is the standard second-order central difference.

2.2.2 Nonlinear Term (Advection) Linearization

To handle the nonlinear term auu_x , we employ a Predictor-Corrector scheme based on the technique of a previous time step prediction, $u_j^{n+1} \approx u_j^n$. The full term is linearized as:

$$auu_x \approx \frac{a}{2} \left[u_j^n \left(\frac{u_{j+1}^{n+1} - u_{j-1}^{n+1}}{2\Delta x} \right) + u_j^n \left(\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} \right) \right] \quad (4)$$

This renders the entire system linear with respect to the unknown terms u^{n+1} .

3 The Final Iterative System

Substituting all approximations into **Equation 1** results in a tridiagonal linear system of equations that must be solved at each time step n :

$$A_j u_{j-1}^{n+1} + B_j u_j^{n+1} + C_j u_{j+1}^{n+1} = R_j^n \quad (5)$$

3.1 Coefficients (Left-Hand Side, u^{n+1} terms)

The coefficients for the unknown vector u^{n+1} are:

- $A_j = -\frac{au_j^n}{4\Delta x} - \frac{c}{2(\Delta x)^2}$
- $B_j = K + \frac{c}{(\Delta x)^2}$
- $C_j = \frac{au_j^n}{4\Delta x} - \frac{c}{2(\Delta x)^2}$

3.2 Right-Hand Side (RHS)

The RHS vector, R_j^n , contains all known values from time steps n and earlier. It is composed of a history term (H_j^n) and a source term (S_j^n): $R_j^n = H_j^n + S_j^n$.

1. **History Term (H_j^n):** This term captures the memory effect of the fractional derivative, depending on all previous time steps.

$$H_j^n = Ku_j^n - K \sum_{k=1}^n b_k (u_j^{n+1-k} - u_j^{n-k})$$

2. **Source Term (S_j^n):** This term captures the discretized spatial terms at time t_n .

$$S_j^n = -\frac{a}{2}u_j^n \left(\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} \right) + \frac{c}{2} \left(\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta x)^2} \right)$$

4 Analysis and Validation

4.1 Stability and Accuracy

4.1.1 Stability

The implicit nature of the Crank-Nicolson part of the scheme, combined with the L1 approximation, ensures **unconditional stability**. This is a crucial feature, as it allows for the use of larger time steps (Δt) without leading to numerical errors growing uncontrollably, which is a common limitation of conditionally stable explicit schemes.

4.1.2 Accuracy

The theoretical accuracy of this scheme is predicted to be $O(\Delta t^{2-\alpha}, \Delta x^2)$.

- $O(\Delta x^2)$: Second-order accuracy in space, resulting from the central difference approximation of the viscous term.
- $O(\Delta t^{2-\alpha})$: Fractional-order accuracy in time, dictated by the combination of the L1 formula and the C-N predictor-corrector approach. For the classical case $\alpha = 1$, this reduces to $O(\Delta t)$, and for $\alpha = 0.5$, it yields $O(\Delta t^{1.5})$.

4.2 Rate of Convergence Verification

Since no simple exact analytical solution exists for the general nonlinear FBE, the standard method for validation is to verify that the numerical solution converges at the predicted theoretical rate. The convergence rate p is calculated using the formula:

$$p = \log_2 \left(\frac{E_1}{E_2} \right)$$

where E_1 and E_2 are the errors between successive grid refinements (e.g., Δt_1 vs. $\Delta t_2 = \Delta t_1/2$).

For a test case with $\alpha = 0.5$, the theoretical temporal order is $p = 2 - 0.5 = 1.5$. Simulation results based on refining the time step show:

- Calculated Ratio $\frac{E_1}{E_2} \approx 2.807$
- Calculated Order $p \approx \log_2(2.807) \approx \mathbf{1.49}$

The close agreement between the calculated order (1.49) and the theoretical order (1.5) confirms the correctness and accuracy of the implemented numerical scheme.

5 Conclusion

The L1/Crank-Nicolson Predictor-Corrector scheme successfully provides a robust and computationally fast method for solving the nonlinear time-fractional Burgers' equation. The core objectives were met:

- The scheme is **unconditionally stable**, allowing for flexible time step selection.
- The **L1 formula** effectively models the non-local memory of the fractional derivative.
- The **Predictor-Corrector linearization** successfully converts the nonlinear PDE into a linear tridiagonal system.
- **Convergence analysis** confirms the scheme's high order of accuracy in both time ($O(\Delta t^{2-\alpha})$) and space ($O(\Delta x^2)$).

The implementation used for the project is available via the public repository: <https://github.com/vedjaw/Fractional-Burger-Equation-Solver>.